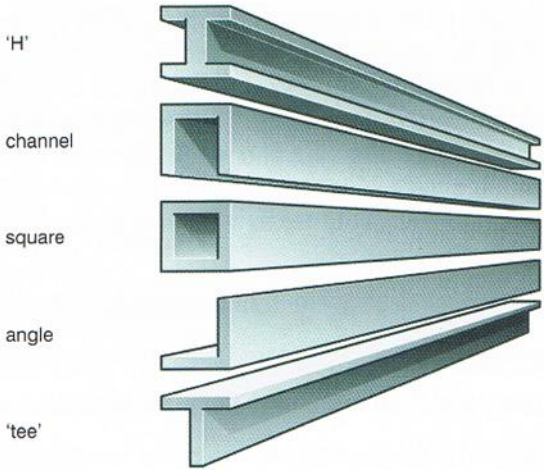
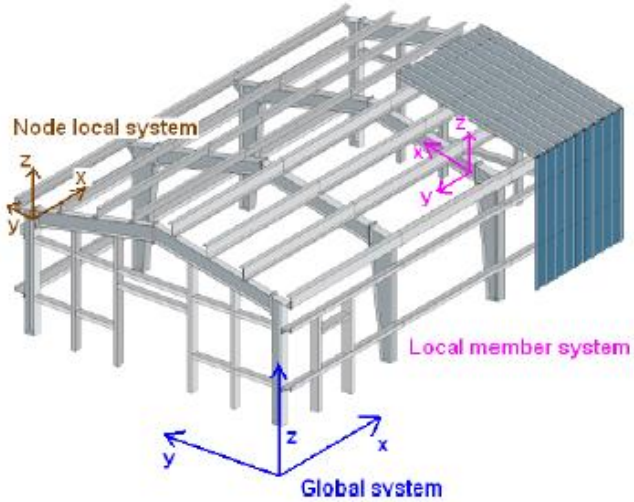
ESA 321
Aerospace Structures

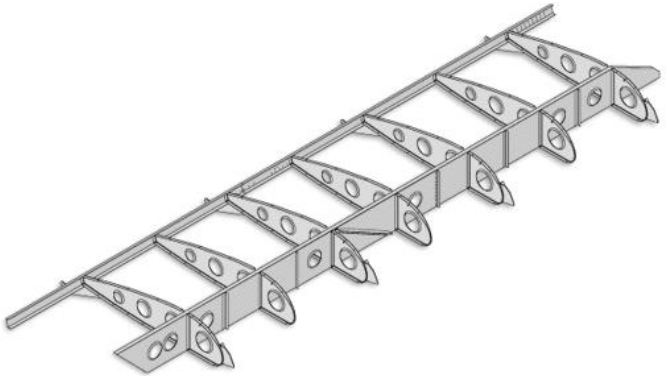
Lecture 3 – Beam & Frame Element



Objectives

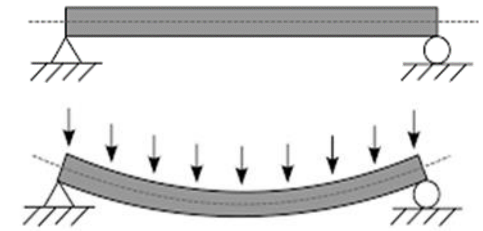
- Able to apply FEM using beam and frame element for solving structural problems.

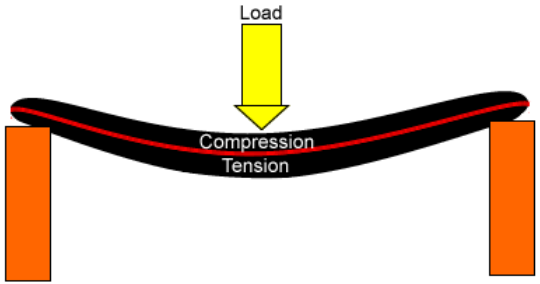




Beam Element

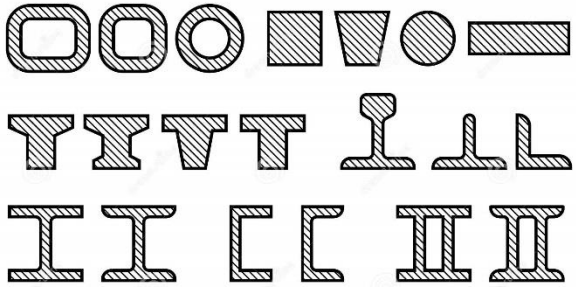
- A beam is a bar-like structural member whose primary function is to **support transverse loading** and carry it to the supports.
- A beam resists **transverse loads** mainly through bending action. Bending produces **compressive** longitudinal stresses in one side of the beam and **tensile** stresses in the other.



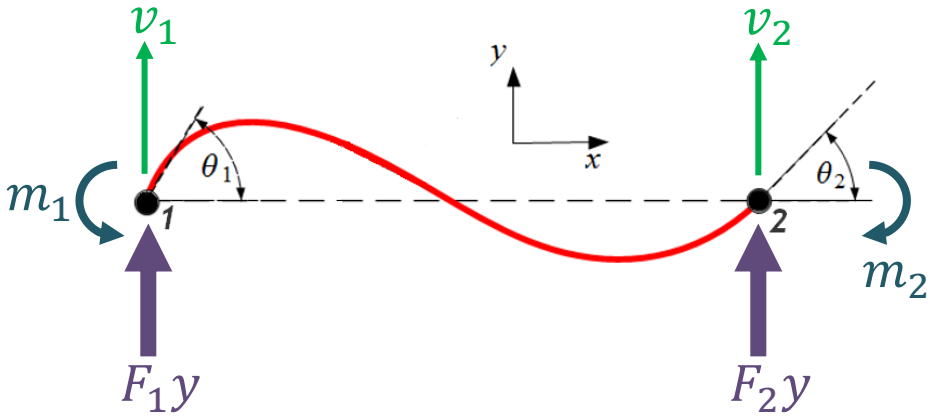


- **A beam element is used to model**
  - A **long** and **slender** structural member that experiences **bending** due to **transverse loading**.
  - Bending deformation is measured as **transverse displacement** and **rotation**.

Beam Element Properties



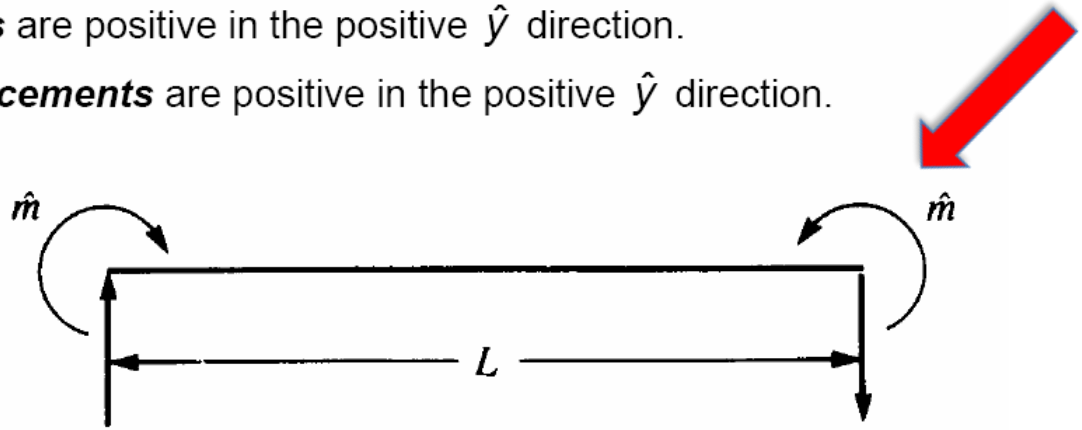
- Beam elements are considered straight and have **constant cross section area**.
- Deform only in the directions **perpendicular** to the axis of the beam.
- Each node of the beam element has **2 dof** (**transverse displacement, v** and **rotation, θ**).
- Beams are only subjected to **transverse forces** **and** **moments**.
- In beam structures, the beams are joined together by **welding** (not by pins or hinges).
- Each beam element has **length, L**, **second area moment, I** and **Elastic modulus, E**.



Beam Element (2)

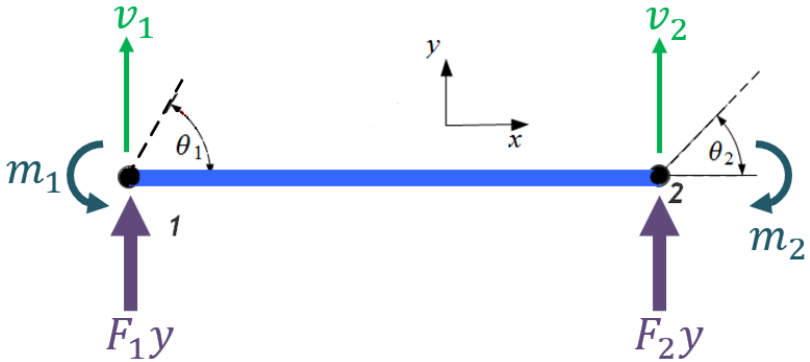
At all nodes, the following sign conventions are used:

1. ***Moments*** are positive in the counterclockwise direction.
2. ***Rotations*** are positive in the counterclockwise direction.
3. ***Forces*** are positive in the positive $\hat{y}$ direction.
4. ***Displacements*** are positive in the positive $\hat{y}$ direction.



Beam Element Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix}=\begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$



$$[K]=\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

At all nodes, the following sign conventions are used:

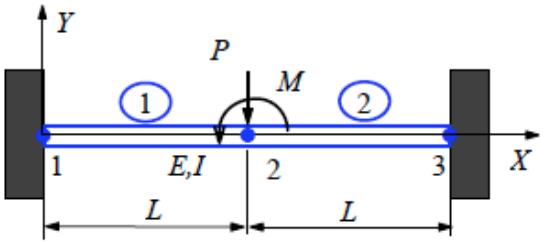
1. ***Moments*** are positive in the counterclockwise direction.
2. ***Rotations*** are positive in the counterclockwise direction.
3. ***Forces*** are positive in the positive $\hat{y}$ direction.
4. ***Displacements*** are positive in the positive $\hat{y}$ direction.

L is the element length
I is the second area moment
E is the modulus of elasticity

Example

Element 1

Element 2



The beam shown above is clamped at the two ends and acted upon by the force P and moment M in the mid-span.

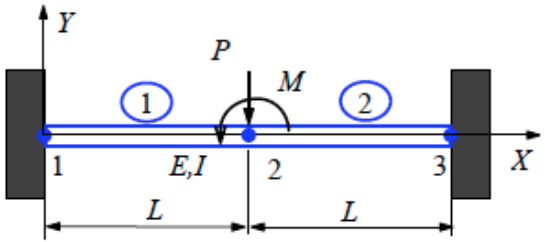
The deflection and rotation at the center node and the reaction forces and moments at the two ends.

$$F_{2Y} = -P, \qquad M_2 = M,$$
$$v_1 = v_3 = \theta_1 = \theta_3 = 0$$

Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$

Example



The beam shown above is clamped at the two ends and acted upon by the force P and moment M in the mid-span.

The deflection and rotation at the center node and the reaction forces and moments at the two ends.

$$F_{2Y} = -P, \qquad M_2 = M,$$
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Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$

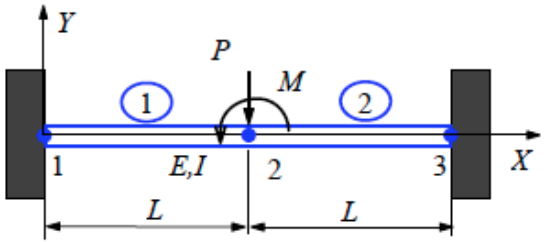
Element 1

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

Example



The beam shown above is clamped at the two ends and acted upon by the force P and moment M in the mid-span.

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$$F_{2Y} = -P, \qquad M_2 = M,$$
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Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$

Element 1

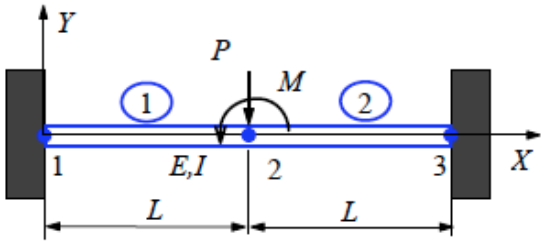
$$\begin{matrix} v_1 & \theta_1 & v_2 & \theta_2 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

$$\begin{matrix} v_2 & \theta_2 & v_3 & \theta_3 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

$$\begin{matrix} v_1 & \theta_1 & v_2 & \theta_2 & v_3 & \theta_3 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}\begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{bmatrix} F_1 y \\ M_1 \\ -P \\ M \\ F_3 y \\ M_3 \end{bmatrix}$$

Example



The beam shown above is clamped at the two ends and acted upon by the force P and moment M in the mid-span.

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Formulation

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Element 1

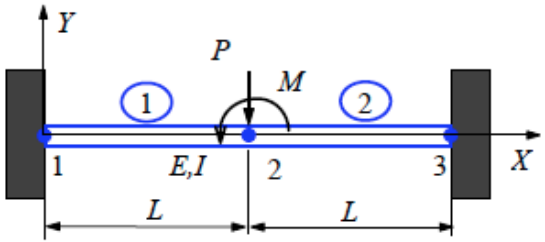
$$\begin{matrix} v_1 & \theta_1 & v_2 & \theta_2 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

$$\begin{matrix} v_2 & \theta_2 & v_3 & \theta_3 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

$$\begin{matrix} v_1 & \theta_1 & v_2 & \theta_2 & v_3 & \theta_3 \end{matrix}$$
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L & 0 & 0 \\ 6L & 4L^2 & -6L & 2L^2 & 0 & 0 \\ -12 & -6L & 24 & 0 & -12 & 6L \\ 6L & 2L^2 & 0 & 8L^2 & -6L & 2L^2 \\ 0 & 0 & -12 & -6L & 12 & -6L \\ 0 & 0 & 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{bmatrix} F_1 y \\ M_1 \\ -P \\ M \\ F_3 y \\ M_3 \end{bmatrix}$$

Example



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Element 1

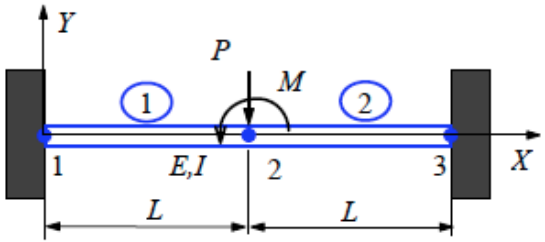
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L & 0 & 0 \\ 6L & 4L^2 & -6L & 2L^2 & 0 & 0 \\ -12 & -6L & 24 & 0 & -12 & 6L \\ 6L & 2L^2 & 0 & 8L^2 & -6L & 2L^2 \\ 0 & 0 & -12 & -6L & 12 & -6L \\ 0 & 0 & 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{bmatrix} F_1y \\ M_1 \\ -P \\ M \\ F_3y \\ M_3 \end{bmatrix}$$

Example



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$$F_{2Y} = -P, \qquad M_2 = M,$$
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Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$

Element 1

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

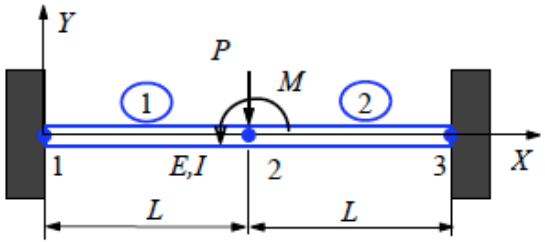
$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L & 0 & 0 \\ 6L & 4L^2 & -6L & 2L^2 & 0 & 0 \\ -12 & -6L & 24 & 0 & -12 & 6L \\ 6L & 2L^2 & 0 & 8L^2 & -6L & 2L^2 \\ 0 & 0 & -12 & -6L & 12 & -6L \\ 0 & 0 & 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{bmatrix} F_1y \\ M_1 \\ -P \\ M \\ F_3y \\ M_3 \end{bmatrix}$$

Reduced to

$$\frac{EI}{L^3}\begin{bmatrix} 24 & 0 \\ 0 & 8L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} -P \\ M \end{Bmatrix}$$

Example



The beam shown above is clamped at the two ends and acted upon by the force P and moment M in the mid-span.

The deflection and rotation at the center node and the reaction forces and moments at the two ends.

$$F_{2Y} = -P, \qquad M_2 = M,$$
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Formulation

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} \\ m_1 \\ F_{2y} \\ m_2 \end{Bmatrix}$$

Element 1

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}$$

Element 2

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{bmatrix} v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{bmatrix}$$

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L & 0 & 0 \\ 6L & 4L^2 & -6L & 2L^2 & 0 & 0 \\ -12 & -6L & 24 & 0 & -12 & 6L \\ 6L & 2L^2 & 0 & 8L^2 & -6L & 2L^2 \\ 0 & 0 & -12 & -6L & 12 & -6L \\ 0 & 0 & 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{bmatrix} F_1y \\ M_1 \\ -P \\ M \\ F_3y \\ M_3 \end{bmatrix}$$

Reduced to

$$\frac{EI}{L^3}\begin{bmatrix} 24 & 0 \\ 0 & 8L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} -P \\ M \end{Bmatrix}$$

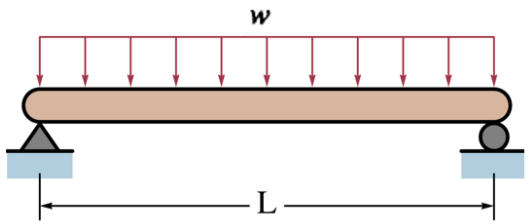
Rearrange

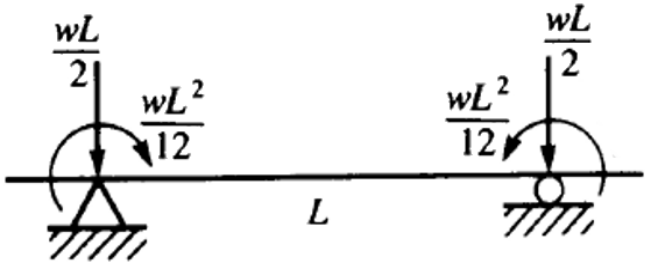
$$\begin{Bmatrix} v_2 \\ \theta_2 \end{Bmatrix} = \frac{L}{24EI}\begin{Bmatrix} -PL^2 \\ 3M \end{Bmatrix}$$

Distributed Load

A load applied across a length or area instead of at one point.

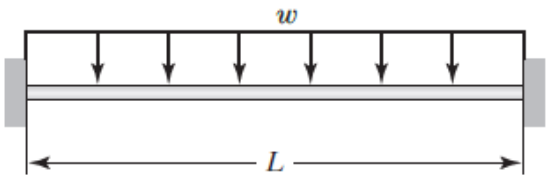
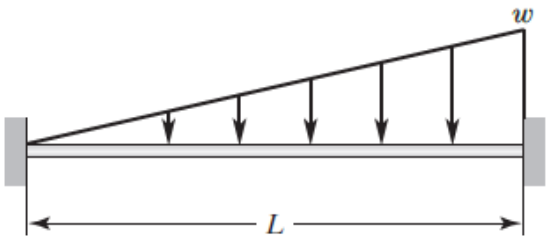
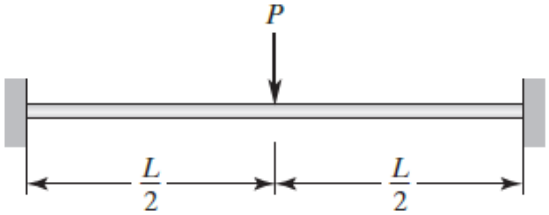
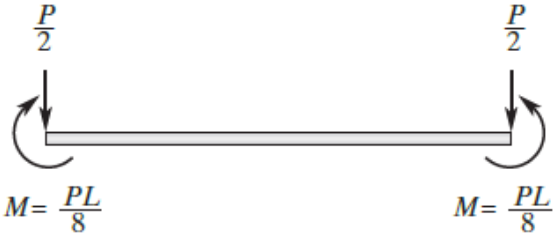
Inn FEM, loads must be applied only at the nodes.





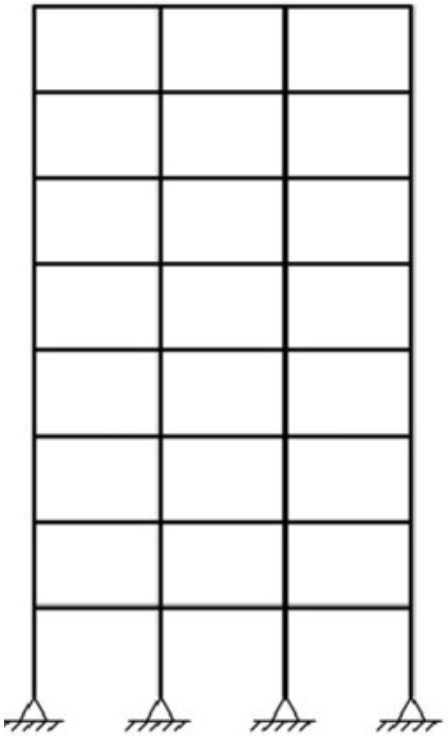
Distributed load must be equated with equivalent concentrated load applied at the element nodes.

TABLE 4.2 Equivalent nodal loading of beams

Loading	Equivalent Nodal Loading
	
	
	

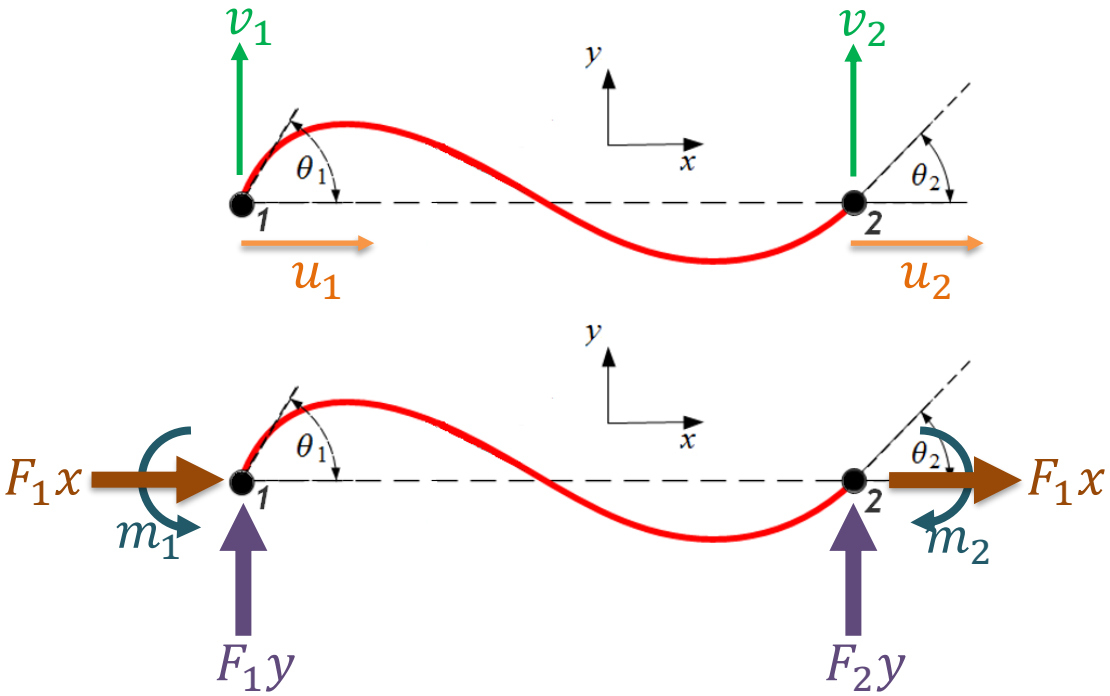
2D Frame Element

- A rigid plane frame is defined as **a series of beam elements rigidly connected to each other.**
- **The angles** made between elements at joints **remained unchanged** after the deformation.
- Moments are transmitted from one element to another at joints.



2D Frame Element Properties

- **Frame elements has 3 dof.**
- **It can be subjected to axial and transverse forces, as well as moments.**
- **Frame elements are applicable for the analysis of skeletal type systems of both *planar frames* (2D frames) and *space frames* (3D frames).**
- **Known generally as the beam element or general beam element in most commercial software.**



2D Frame Element

- Bar + beam

$$\mathbf{k}_e^{truss} = \begin{bmatrix} \frac{AE}{2a} & 0 & 0 & -\frac{AE}{2a} & 0 & 0 \\ & 0 & 0 & 0 & 0 & 0 \\ & & 0 & 0 & 0 & 0 \\ & & & \frac{AE}{2a} & 0 & 0 \\ & sy. & & & 0 & 0 \\ & & & & & 0 \end{bmatrix} + \mathbf{k}_e^{beam} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ & \frac{3EI_z}{2a^3} & \frac{3EI_z}{2a^2} & 0 & -\frac{3EI_z}{2a^3} & \frac{3EI_z}{2a^2} \\ & & \frac{2EI_z}{a} & 0 & -\frac{3EI_z}{2a^2} & \frac{EI_z}{a} \\ & & & 0 & 0 & 0 \\ & sy. & & & \frac{3EI_z}{2a^3} & -\frac{3EI_z}{2a^2} \\ & & & & & \frac{2EI_z}{a} \end{bmatrix}$$

$l=2a$

$$\mathbf{k}_e = \begin{bmatrix} \frac{AE}{2a} & 0 & 0 & -\frac{AE}{2a} & 0 & 0 \\ & \frac{3EI_z}{2a^3} & \frac{3EI_z}{2a^2} & 0 & -\frac{3EI_z}{2a^3} & \frac{3EI_z}{2a^2} \\ & & \frac{2EI_z}{a} & 0 & -\frac{3EI_z}{2a^2} & \frac{EI_z}{a} \\ & & & \frac{AE}{2a} & 0 & 0 \\ & sy. & & & \frac{3EI_z}{2a^3} & -\frac{3EI_z}{2a^2} \\ & & & & & \frac{2EI_z}{a} \end{bmatrix}$$

$$\begin{Bmatrix} u_1 \\ v_1 \\ \theta_{z1} \\ u_2 \\ v_2 \\ \theta_{z2} \end{Bmatrix}$$

Bar

Beam

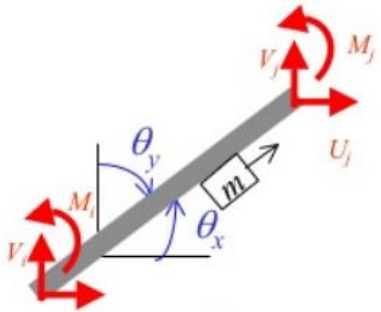
Equations in local reference frame

2D Frame Element

$$\sigma = E\varepsilon$$

Transform from local to global reference frame

$$\mathbf{K}_e = \mathbf{T}^T \mathbf{k}_e \mathbf{T}$$



Global

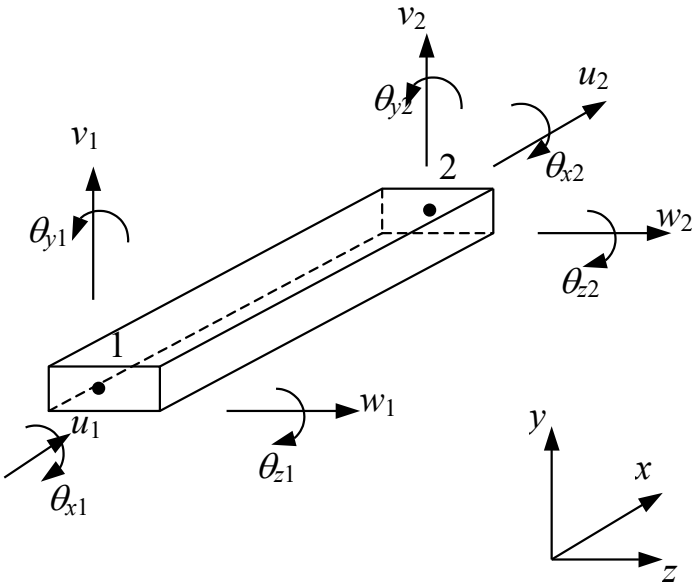
$$\mathbf{K}_e = \frac{E}{L}\begin{bmatrix}
A\cos^2\theta + \frac{12I\sin^2\theta}{L^2} & \sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & \frac{-6I\sin\theta}{L} & -\left(A\cos^2\theta + \frac{12I\sin^2\theta}{L^2}\right) & -\sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & \frac{-6I\sin\theta}{L} \\
\sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & A\sin^2\theta + \frac{12I\cos^2\theta}{L^2} & \frac{6I\cos\theta}{L} & -\sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & -\left(A\sin^2\theta + \frac{12I\cos^2\theta}{L^2}\right) & \frac{6I\cos\theta}{L} \\
\frac{-6I\sin\theta}{L} & \frac{6I\cos\theta}{L} & 4I & \frac{6I\sin\theta}{L} & -\frac{6I\cos\theta}{L} & 2I \\
-\left(A\cos^2\theta + \frac{12I\sin^2\theta}{L^2}\right) & -\sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & \frac{6I\sin\theta}{L} & A\cos^2\theta + \frac{12I\sin^2\theta}{L^2} & \sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & \frac{6I\sin\theta}{L} \\
-\sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & -\left(A\sin^2\theta + \frac{12I\cos^2\theta}{L^2}\right) & -\frac{6I\cos\theta}{L} & \sin\theta\cos\theta\left(A - \frac{12I}{L^2}\right) & A\sin^2\theta + \frac{12I\cos^2\theta}{L^2} & -\frac{6I\cos\theta}{L} \\
\frac{-6I\sin\theta}{L} & \frac{6I\cos\theta}{L} & 2I & \frac{6I\sin\theta}{L} & -\frac{6I\cos\theta}{L} & 4I
\end{bmatrix}$$

$$\lambda_x = \cos\theta_x$$

$$\lambda_y = \cos\theta_y$$

3D Frame Element

- Displacement and forces in 3 axes, torsion.



$$\mathbf{d}_e = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \\ d_9 \\ d_{10} \\ d_{11} \\ d_{12} \end{Bmatrix} = \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \theta_{x1} \\ \theta_{y1} \\ \theta_{z1} \\ u_2 \\ v_2 \\ w_2 \\ \theta_{x2} \\ \theta_{y2} \\ \theta_{z2} \end{Bmatrix}$$

dof of at node 1

dof of at node 2

3D Frame Element

- Truss + beam (local coordinate)

$$\mathbf{K}_e = \mathbf{T}^T \mathbf{k}_e \mathbf{T}$$
$$\mathbf{M}_e = \mathbf{T}^T \mathbf{m}_e \mathbf{T}$$
$$\mathbf{F}_e = \mathbf{T}^T \mathbf{f}_e$$

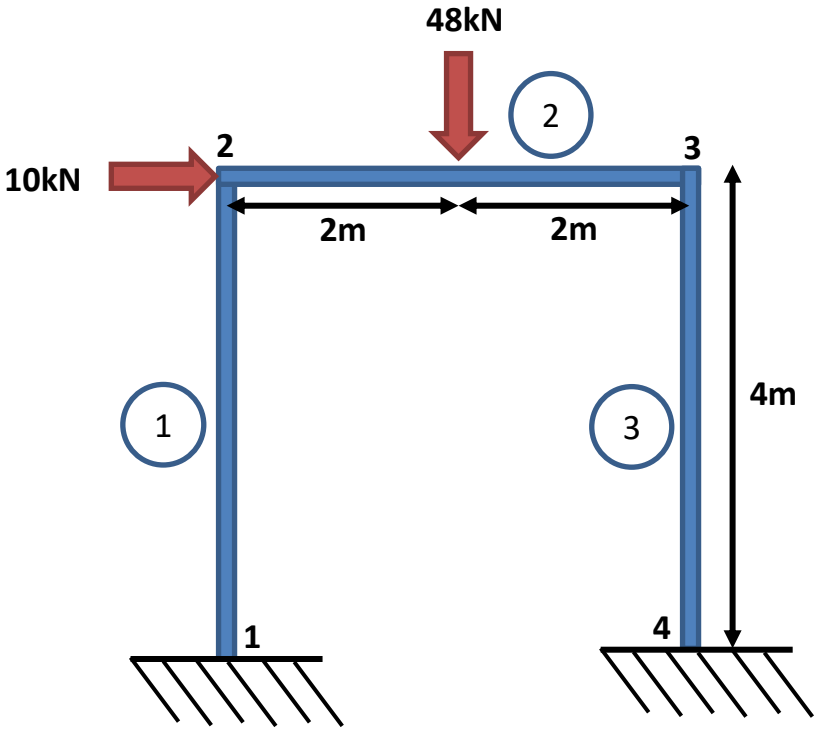
$$\mathbf{k}_e = \begin{array}{c} \begin{matrix} u_1 & v_1 & w_1 & \theta_{x1} & \theta_{y1} & \theta_{z1} & u_2 & v_2 & w_2 & \theta_{x2} & \theta_{y2} & \theta_{z2} \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{matrix} \\ \begin{bmatrix} \frac{AE}{2a} & 0 & 0 & 0 & 0 & 0 & \frac{-AE}{2a} & 0 & 0 & 0 & 0 & 0 \\ & \frac{3EI_z}{2a^3} & 0 & 0 & 0 & \frac{3EI_z}{2a^2} & 0 & \frac{-3EI_z}{2a^3} & 0 & 0 & 0 & \frac{3EI_z}{2a^2} \\ & & \frac{3EI_y}{2a^3} & 0 & \frac{-3EI_y}{2a^2} & 0 & 0 & 0 & \frac{-3EI_y}{2a^3} & 0 & \frac{-3EI_y}{2a^2} & 0 \\ & & & \frac{GJ}{2a} & 0 & 0 & 0 & 0 & 0 & \frac{-GJ}{2a} & 0 & 0 \\ & & & & \frac{2EI_y}{a} & 0 & 0 & 0 & \frac{3EI_y}{2a^2} & 0 & \frac{EI_y}{a} & 0 \\ & & & & & \frac{2EI_z}{a} & 0 & \frac{-3EI_z}{2a^2} & 0 & 0 & 0 & \frac{EI_z}{a} \\ & & & & & & \frac{AE}{2a} & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & \frac{3EI_z}{2a^3} & 0 & 0 & 0 & \frac{-3EI_z}{2a^2} \\ & & & & & & & & \frac{3EI_y}{2a^3} & 0 & \frac{3EI_y}{2a^2} & 0 \\ & & & sy. & & & & & & \frac{GJ}{2a} & 0 & 0 \\ & & & & & & & & & & \frac{2EI_y}{a} & 0 \\ & & & & & & & & & & & \frac{2EI_z}{a} \end{bmatrix} \end{array}$$

$l=2a$

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{T}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{T}_3 \end{bmatrix}$$

$$\mathbf{T}_3 = \begin{bmatrix} l_x & m_x & n_x \\ l_y & m_y & n_y \\ l_z & m_z & n_z \end{bmatrix}$$

Practice –Frame Element



Discretize the structure into 3 frame elements and find:

1) **Total forces and moment acting on node 2 and 3**
2) **Global stiffness matrix**
3) **Displacement of node 2 and node 3**

Parameter	Value
I	1.33×10^{-4} m^4
A	0.1 m^2
E	20 GPa
Same for all element	

Thank you